Total No. of Questions : 10]

SEAT No. :

P185

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B.E. (Computer Engg.)

DATA MINING AND WAREHOUSING

(2015 Pattern) (Semester - I) (410244 D) (Elective-I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) How to compute dissimilarity for binary attributes with examples. **[4]**

b) Explain Knowledge discovery from data or KDD. **[6]**

OR

Q2) a) Explain types of attributes with examples. **[4]**

b) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 16, 17, 20, 22, 25, 25, 28, 30, 35, 40, 42, 42, 45, 50, 52, 55, 60, 60, 62, 62, 65, 70 Partition them into three bins by each of the following methods.

i) Equal-frequency partitioning

ii) Equal-width partitioning

Use *smoothing by bin means* to smooth the above data. **[6]**

Q3) a) Explain any two data reduction techniques. **[4]**

b) Compare Online Analytical Processing (OLAP) and Online Transaction processing (OLTP). **[6]**

OR

Q4) a) Explain following. **[5]**

i) Minskowski Distance

ii) Euclidean distance

P.T.O.

b) Briefly compare the following concepts. You may use an example to explain your point(s). **[6]**

Snowflake schema, fact constellation, star schema.

Q5) a) Consider the market basket transactions shown below: **[6]**

Transaction ID	Items bought
T1	{M, A, B, D}
T2	{A, D, C, B, F}
T3	{A, C, B, F}
T4	{A, B, D}

Assuming the minimum support of 50% and minimum confidence of 80%

i) Find all frequent itemsets using Apriori algorithm.

ii) Find all association rules using Apriori algorithm.

b) Explain mining Multilevel association rules. What is Uniform support? **[6]**

c) Define support and confidence for an association rule. **[4]**

OR

Q6) a) A database has five transactions. Let $min\_sup = 60\%$ *and* $min\_conf = 80\%$ **[6]**

TID	items _bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

i) Find all frequent itemsets using FP-growth

ii) List all the strong association rules (with support s and confidence c)

b) Explain the following terms: **[6]**

i) Constraint based rule mining

ii) Closed and maximal frequent itemsets

c) What do you mean by frequent item set, Closed item set? Explain with example. **[4]**

Q7) a) Define Classification and Prediction. Explain decision tree based Classification method with suitable example. **[8]**

b) Describe K-Nearest Neighbor classifiers with suitable example. **[6]**

c) Write short note on Rule Induction Using a Sequential Covering Algorithm. **[4]**

OR

Q8) a) Explain the following **[8]**

i) Gini index

ii) Gain ratio

iii) Information gain

b) Differentiate between Supervised and unsupervised Learning **[6]**

c) What are Bayesian classifiers? **[4]**

Q9) a) Explain following with example **[8]**

i) Accuracy

ii) Error Rate

iii) Precision

iv) Recall

b) Describe following. **[8]**

i) Multiclass classification

ii) Reinforcement learning

OR

Q10) a) Explain in detail following techniques to evaluate the accuracy of a Classifier. **[8]**

i) Holdout method

ii) Cross validation

b) Explain following **[8]**

i) Systematic learning

ii) Wholistic learning